Exchange Rate Disconnect with Disagreeing Arbitrageurs

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The big picture: exchange rate disconnect puzzles

- **Macro-FX disconnect**: Exchange rates are
  - inconsistent with complete markets: Backus-Smith (1993)
  - hard to predict: Meese-Rogoff (1983)
  - inconsistent with UIP: Hansen-Hodrick (1980); Fama (1984)
- Leading *fix*: segmented asset market with a **shock to the UIP condition**
  Gabaix & Maggiori (2015); Itskhoki & Mukhin (2021)

$$\mathbb{E}_t \Delta \hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t = \hat{\psi}_t + \text{conventional terms.}$$

  - Interpretations of ψ_t: risk capacity shock, noise trader shock, etc
- *Recent theoretical work* fixes FX-disconnect via **exogenous** UIP shocks
- *Recent empirical evidence* suggests that FX is actually *connected* to macro
  Hassan-Mertens-Zhang (2016); Lilley-Maggiori-Neiman-Schreger (2022); Engel-Wu (2024); Kekre-Lenel (2024); Chahrour et al. (2025); Stavrakeva-Tang (2026)

Alternative approach: endogenous UIP wedge from disagreement and wealth shares

Exchange rate disagreement facts

- Disagreement about the future exchange rate is large and persistent
- It tends to increase during risk episodes: e.g., Global Financial Crisis, and COVID-19
- But it never closes: even during the calmest periods, disagreement is still large

Is this just a sideshow — or can disagreement itself be the structural source of the UIP wedge?

▸ Disagreement over time

This paper

- Builds on the success of recent international macro/finance models
  Segmented asset market with international arbitrageurs; home-bias in consumption; imperfect pass-through
- Introduces international arbitrageurs with *heterogeneous beliefs (disagreement)*
  - Disagreement $\Rightarrow$ different portfolio positions $\Rightarrow$ *different wealth shares*

$\Rightarrow$ a novel UIP wedge: **wealth-weighted** belief $=$ **consensus** $+$ **covariance**

$$\mathbb{E}_t \Delta \hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t = - \underbrace{Cov_t^c \left(E_t^i \left[\hat{e}_{t+1} \right], \omega_t^i \right)}_{\text{beliefs x wealth share}} + \text{conventional terms}$$

- The wedge is **endogenous** to macro shocks: no exogenous wedge process
  - FX responses are amplified; this mechanism is sufficient to match FX-macro moments
- Calibrated model **quantitatively** matches important FX-macro moments

Disagreement–wealth mechanism

1. **Macro news moves the exchange rate**
   Weaker U.S. demand depreciates the dollar

2. **Disagreement means different positions**
   Foreign optimists are long Foreign currency; pessimists are short

3. **FX returns redistribute wealth**
   The surprise depreciation raises these optimists' wealth share

4. **Wealth changes whose beliefs matter for prices**
   Foreign optimists carry more weight, amplifying the depreciation

A macro shock generates an endogenous UIP wedge
It is shown that this leads to amplification and FX disconnect **without UIP shocks**

Recent literature: what drives exchange rates?

- **Itskhoki & Mukhin (2021)**: UIP shock view
  - Exchange rates dynamics $\approx$ exogenous UIP shock
- **Kekre & Lenel (2024)**: discount factor (long yield) view
  - Challenge IM: "Exchange rates have structure"
  - Long yield differentials ($\approx$ discount factor shocks) drive exchange rates
  - But still need a UIP shock to match some moments

This paper: structural origin of the UIP "shock" both sides still need

The key is a large **UIP wedge**: it does not need to be an exogenous shock

▸ full literature

Plan of the presentation

Introduction

Model

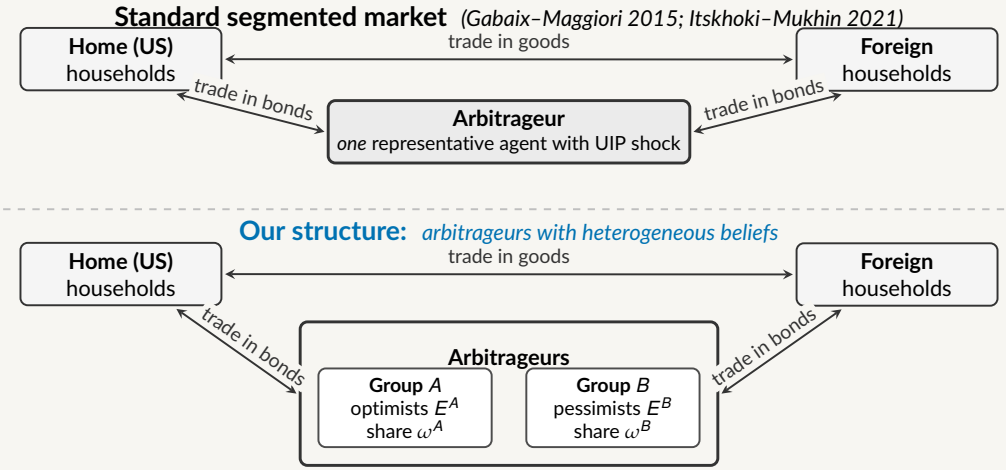
Analytical Insights

Quantitative Analysis

Conclusion

Model

Model overview: segmented market with heterogeneous arbitrageurs



Home households

The representative Home household's problem is given by

$$\max_{\{c_{t+j}, b_{t+j}\}_{j=0}^{\infty}} \mathbb{E}_t \sum_{j=0}^{\infty} \beta_{t+j} \left[\frac{c_{t+j}^{1-\sigma}}{1-\sigma} \right]$$

subject to

$$c_t + \frac{b_t}{1+r_t} = p_t y_t + b_{t-1} + \pi_t + \pi_t^a,$$

where domestic and import goods with home bias

$$c_t = \left[(1-\gamma)^{\frac{1}{\theta}} c_{Ht}^{\frac{\theta-1}{\theta}} + \gamma^{\frac{1}{\theta}} c_{Ft}^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}$$

Symmetric for Foreign: discount factor β_t^*, endowment y_t^*, bond b_t^*, and consumption c_t^*,

Firms exist only to break perfect exchange-rate pass-through, with degree χ: nests LOP ▸ pricing-to-market

Proposed by Alessandria (2009); Kekre-Lenel (2024)

Arbitrageurs: setup

- A continuum of price-taking arbitrageurs:
  - two groups $i \in \{A, B\}$; beliefs $E_t^i[\cdot]$; masses $\eta_A + \eta_B = 1$
- Self financing carry trade
  - Borrow in Home bond, lend in Foreign bond (and vice versa); amount matched exactly
- Myopic mean–variance objective over next-period wealth
  - Risk tolerance increases with wealth: approximation of CRRA; proposed by Kekre-Lenel (2026)
- Finite lives:
  - Exit with probability ξ each period; replaced by new entrants with wealth W_0^i
- Net profits rebated to households

Arbitrageurs' problem

- Choose the current carry position $n_t^i \left(= \frac{e_t b_t^{*i}}{1+r_t^*} = -\frac{b_t^i}{1+r_t} \right)$:

$$\max_{n_t^i} \left\{ E_t^i[w_{t+1}^i] - \frac{\nu}{2(1+r_t)w_t^i} \operatorname{Var}_t^i[w_{t+1}^i] \right\}, \qquad \nu > 0$$

subject to wealth dynamics before exit, with carry return R_{t+1}^x:

$$w_{t+1}^i = (1+r_t)w_t^i + n_t^i R_{t+1}^x, \qquad R_{t+1}^x \equiv \frac{e_{t+1}}{e_t}(1+r_t^*) - (1+r_t)$$

- W_t^i, N_t^i: wealth and carry position per group after exit and entry

$$W_{t+1}^i = (1-\xi)\left((1+r_t)W_t^i + N_t^i R_{t+1}^x\right) + \xi W_0^i$$

- FOC with log-return approximation:

$$\frac{N_t^i}{W_t^i} = \frac{E_t^i\left[\Delta \log e_{t+1}\right] + \log\left(1+r_t^*\right) - \log\left(1+r_t\right)}{\nu \operatorname{Var}_t^i\left[\Delta \log e_{t+1}\right]}.$$

Equilibrium and Solution

Equilibrium

- Four markets clear: Home and Foreign goods, Home and Foreign bond markets
- Four exogenous variables: $y_t, y_t^*, \beta_t, \beta_t^*$

Approximation protocol (Borovička–Hansen–Sargent 2023; as in Itskhoki–Mukhin 2021)

- Small noise: scale shocks by δ, let $\nu(\delta)\,\delta^2 \to \Gamma$; linearize around the deterministic SS
- Portfolio choice is bounded at first order

Steady-state

- Permanent disagreement: $d_{ss} \equiv E^A[\log e] - E^B[\log e] > 0$
- Endowments are set so that all prices are set to 1

Analytical Insights

Toward a simplified equilibrium system

Keep permanent disagreement; let macro shocks redistribute wealth.

- Large financial sector: bond-supply terms vanish: common trick e.g., Itskhoki-Mukhin (2021)
- Symmetric steady state: $\beta = \beta^*$, $\omega^A = \omega^B = \frac{1}{2}$
- AR(1) fundamentals
- Symmetric, constant disagreement: $E_t^i[\log e_{t+1}] = \mathbb{E}_t[\log e_{t+1}] + E^i[\log e]$
  - Restriction: $E^A[\log e] + E^B[\log e] = 0 (= \mathbb{E}[\log e])$
  - Group A expects Foreign appreciation: $d_{ss} = E^A[\log e] - E^B[\log e] > 0$

Average beliefs are rational. No belief shocks or exogenous UIP shock

$\hat{e} \uparrow$: Home depreciation. Group A is optimistic about Foreign returns

▸ belief formation

These reduce the system to a **three-equation** system

Analytically show that FX-macro moments can be matched without an UIP shock

Bond-market clearing: modified UIP

Portfolio demand scales with wealth: prices reflect wealth-weighted beliefs

$$\underbrace{\mathbb{E}_t \Delta \hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t}_{\text{expected excess return on Foreign bonds}} = -d_{ss} \hat{\omega}_t^A$$

$\hat{\omega}_t^A$: deviation of optimists' wealth share from $\frac{1}{2}$; $d_{ss} = E^A[\log e] - E^B[\log e]$: ss disagreement

- More wealth held by Foreign optimists raises demand for Foreign bonds
- Market clearing requires a lower expected excess return

Households & goods markets: characterizing the interest differential

Goods market clearing

$$\hat{c}_t - \hat{c}_t^* = \phi_y(\hat{y}_t - \hat{y}_t^*) - \chi\phi_e\hat{e}_t$$

Relative Euler equation

$$\hat{r}_t - \hat{r}_t^* = \sigma\mathbb{E}_t[\Delta\hat{c}_{t+1} - \Delta\hat{c}_{t+1}^*] - (\hat{\beta}_t - \hat{\beta}_t^*)$$

Combining the two:

$$\hat{r}_t - \hat{r}_t^* = \hat{\imath}_t^f - S\,\mathbb{E}_t\Delta\hat{e}_{t+1}, \qquad S = \sigma\chi\phi_e,$$

$$\hat{\imath}_t^f = -(\hat{\beta}_t - \hat{\beta}_t^*) - \sigma\phi_y(1-\rho_y)(\hat{y}_t - \hat{y}_t^*)$$

Use the modified UIP to eliminate $\mathbb{E}_t\Delta\hat{e}_{t+1}$:

$$\hat{r}_t - \hat{r}_t^* = \underbrace{\frac{\hat{\imath}_t^f}{1+S}}_{\text{fundamental component}} + \underbrace{\frac{S}{1+S}d_{ss}\hat{\omega}_t^A}_{\text{disagreement–wealth component}}$$

The simplified equilibrium

$\hat{\omega}_t^A$: optimists' wealth share; $d_{ss} > 0$: permanent disagreement.

Three-equation system

$$\hat{r}_t - \hat{r}_t^* = \frac{\hat{\imath}_t^f}{1+S} + \frac{S}{1+S} d_{ss} \hat{\omega}_t^A \qquad \text{Real side + UIP}$$

$$\mathbb{E}_t \Delta \hat{e}_{t+1} + \hat{r}_t^* - \hat{r}_t = -d_{ss} \hat{\omega}_t^A \qquad \text{Bond clearing}$$

$$\hat{\omega}_{t+1}^A = \lambda \hat{\omega}_t^A + b_w \varepsilon_{t+1}^e \qquad \text{Wealth revaluation}$$

$\varepsilon_{t+1}^e \equiv \hat{e}_{t+1} - \mathbb{E}_t \hat{e}_{t+1}$ is determined inside this system.

$0 < \lambda < 1$: wealth persistence. $b_w > 0$: exposure to FX news

Disagreement off: standard model with macro shocks

Drop the entire heterogeneous arbitrageur block

Interest-rate differential

$$\hat{r}_t - \hat{r}_t^* = \frac{\hat{\imath}_t^f}{1+S}$$

$$\hat{\imath}_t^f = -(\hat{\beta}_t - \hat{\beta}_t^*) - \sigma\phi_y(1-\rho_y)(\hat{y}_t - \hat{y}_t^*)$$

Standard UIP

$$\mathbb{E}_t \Delta\hat{e}_{t+1} = \hat{r}_t - \hat{r}_t^*$$

$$\left[\hat{e}_t = -\tfrac{1}{1+S}\mathbb{E}_t \textstyle\sum_{j=0}^{\infty} \hat{\imath}_{t+j}^f \quad \text{(stationary solution)}\right]$$

Example: a Home saving shock

- Home interest rates fall and the Home currency depreciates.
- As the shock fades, the currency appreciates back.

UIP holds and impossible to match FX-disconnect moments

Disagreement–wealth amplifier: the feedback loop

- Exogenous macro shocks $\hat{\imath}_t^f$ drive $\hat{r}_t - \hat{r}_t^*$:

$$\hat{r}_t - \hat{r}_t^* = \frac{\hat{\imath}_t^f}{1+S} + \frac{S}{1+S}\, d_{ss}\hat{\omega}_t^A$$

- UIP prices the interest differential into expected FX changes:

$$\mathbb{E}_t \Delta \hat{e}_{t+1} = (\hat{r}_t - \hat{r}_t^*) - d_{ss}\hat{\omega}_t^A$$

$$\left[\hat{e}_t = -\mathbb{E}_t \textstyle\sum_{j=0}^{\infty} (\hat{r}_{t+j} - \hat{r}_{t+j}^*) + \mathbb{E}_t \sum_{j=0}^{\infty} d_{ss}\hat{\omega}_{t+j}^A \quad \text{(stationary solution)}\right]$$

- FX surprises revalue wealth, changing $\hat{\omega}_t^A$:

$$\hat{\omega}_t^A = \lambda \hat{\omega}_{t-1}^A + b_w \varepsilon_t^e$$

Changes in wealth share further reinforce the fundamental FX responses

Disagreement–wealth amplifier: the fixed point

Define equilibrium exchange-rate innovations

- Disagreement off (fundamental): $\varepsilon^{e,f}_{t+1} = e^f_{t+1} - \mathbb{E}_t e^f_{t+1}$
- Disagreement on: $\varepsilon^{e}_{t+1} = e_{t+1} - \mathbb{E}_t e_{t+1}$

Disagreement–wealth amplifier

$$\varepsilon^{e}_{t+1} = \Xi\, \varepsilon^{e,f}_{t+1}, \qquad \Xi = \frac{1}{1-g} > 1$$

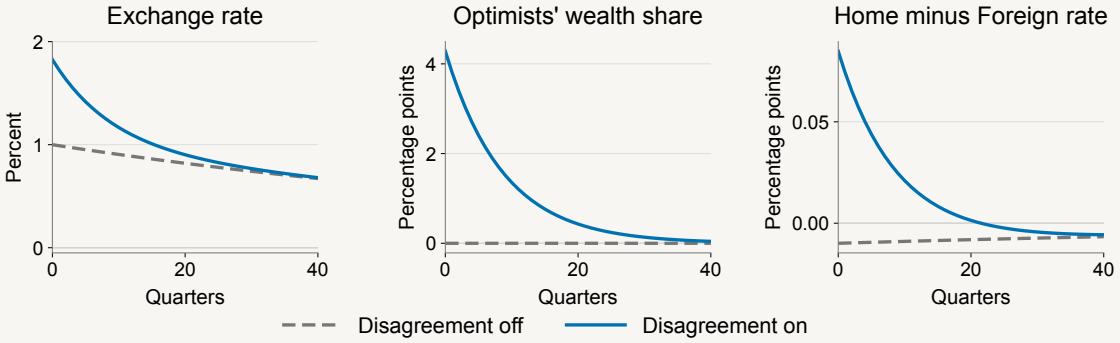
with

$$g = \underbrace{b_w}_{\text{wealth revaluation}} \underbrace{\frac{d_{ss}}{1+S}}_{\text{effect on pricing}} \underbrace{\frac{1}{1-\lambda}}_{\text{persistence}}$$

With disagreement, larger FX moves redistribute wealth and reinforce the price response.

A Home demand shock: numerical example

Higher discount factor: more saving, Home depreciation.



Same macro shock: disagreement amplifies the FX response

Shock scaled to a 1% depreciation with disagreement off.

Exchange-rate disconnect **without UIP shock**: volatility & persistence

No UIP shocks. As fundamentals and wealth become persistent: $\rho_x, \lambda \to 1$.

Fact	Model limit
Persistence FX behaves nearly like a random walk. Meese–Rogoff (1983)	$\text{Corr}(\Delta\hat{e}_t, \Delta\hat{e}_{t-1}) \longrightarrow 0$
Volatility relative to interest rates Large FX moves; small rate movements. Itskhoki–Mukhin (2021)	$\dfrac{\text{sd}(\hat{r}_t - \hat{r}_t^*)}{\text{sd}(\Delta\hat{e}_t)} \longrightarrow 0$

- When states become persistent
  - FX becomes near random-walk
  - Relative volatility becomes arbitrarily small
- Fama: next slide. ▸ Backus–Smith (appendix)

Exchange-rate disconnect **without UIP shock**: Fama puzzle

High-interest-rate currencies often appreciate; UIP predicts depreciation.

Fama (1984)

$$\underbrace{\Delta \hat{e}_{t+1}}_{\text{next-period depreciation}} = \alpha + \beta_F \underbrace{(\hat{r}_t - \hat{r}_t^*)}_{\text{current interest differential}} + u_{t+1}$$

β_F: regression slope. R_F^2: share of FX variation explained.

UIP benchmark: $\beta_F = 1$ with high R_F^2

Model, without UIP shocks: Limit: $\rho_x, \lambda \to 1$ with $\Xi > 1$ fixed and $(1-\rho_x)/(1-\lambda) \to 0$.

$$\underbrace{\beta_F \longrightarrow -\frac{1}{S}}_{\text{high-rate currency appreciates}} \quad , \quad \underbrace{R_F^2 \longrightarrow 0}_{\text{little predictive power}}$$

Matching important FX disconnect moments doesn't require an exogenous UIP shock!
These work under certain limits; do they work quantitatively?

Quantitative Analysis

Baseline: full-information rational expectations (FIRE) benchmark

- Transitory and permanent macro shocks: only the sum is observed
- Disagreement arises from disagreement about permanent components of them
  - Average: rational filtering (FIRE)
  - Time-varying disagreement (AR(1) with constant): ▸ Disagreement over time

$$E_t^A[\tilde{x}_t] - E_t^B[\tilde{x}_t] = (1-\varphi_x)\underbrace{(E^A[x] - E^B[x])}_{\text{ss disagreement}} + \varphi_x\left(E_{t-1}^A[\tilde{x}_{t-1}] - E_{t-1}^B[\tilde{x}_{t-1}]\right) + \epsilon_t^{d_x}$$

  - Arises from idiosyncratic interpretation setting as in Caballero-Simsek (2022) ▸ Belief setup & Proposition 1 (appendix)
  - Private information + price learning leads to similar disagreement dynamics
- Keep nonlinear cross-terms regarding disagreement $(d_t + d_{ss}) * z_t$
- **No UIP shock**

Calibration strategy

Calibrate shocks to match the level of volatility: Kekre-Lenel (2024) approach.

- Data: quarterly, 1991Q2–2020Q1
  - Home = U.S.
  - Foreign = Fed advanced-foreign-economy currency countries: Australia, Canada, Switzerland, the Euro Area, U.K., Japan, and Sweden
    Matched with the dollar forecast in the Blue Chip Financial Forecasts data
- Fix standard parameters: openness, risk aversion, discount factor, etc.
- Calibrate shock processes $(\hat{\beta}_t, \hat{\beta}^*_t, \hat{y}_t, \hat{y}^*_t)$ to match
  - Per-capita GDP and 10 year yield persistence and volatility
  - Relative consumption and 10 year yield cross-country correlation
- Disagreement block: FX disagreement persistence, ss level, and shock scale

▸ FIRE calibration fit ▸ external parameters

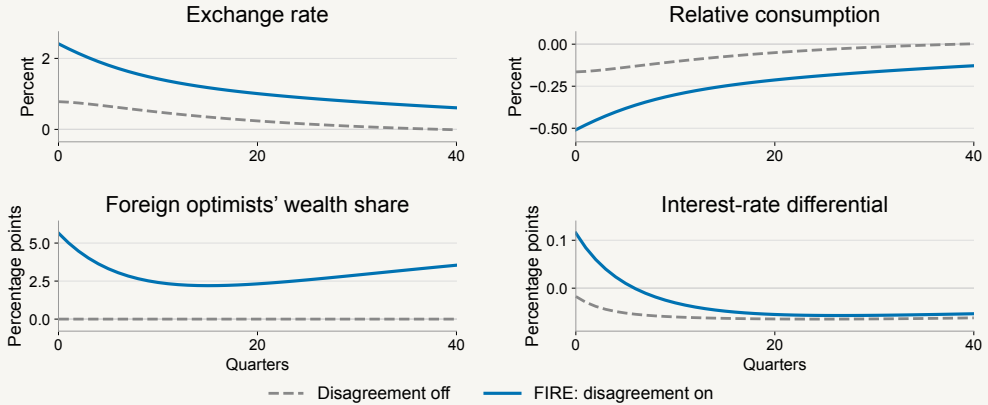
Calibration result overview

- Highly persistent demand and supply shocks: $\rho_\beta \approx 0.99$, $\rho_y \approx 0.987$
  Reflect persistence of long yield and output: echoes Kekre-Lenel (2024)
- Almost exactly matched disagreement block moments: $\rho_d \approx 0.66$, $d_{ss} \approx 0.043$, $\sigma_d \approx 0.0058$
- Positively correlated Home and Foreign consumption: data ≈ 0.63, model ≈ 0.58
  - Recent models compromise: Kekre-Lenel (2024): -0.54; Candian-De Leo (2025) -0.06

The model fits the moment targets well, without any UIP shock. ▸ FIRE calibration fit

FIRE benchmark: disagreement amplifies the same macro shock

Hold macro shocks and risk-bearing capacity fixed. Drop disagreement and heterogeneity.



Impact amplification of exchange rate: 3 times as large

Home saving shock. Consumption and rate differential: Home minus Foreign.

▸ supply shocks ▸ disagreement shocks

FIRE benchmark: untargeted FX-disconnect moments

Hold macro shocks and risk-bearing capacity fixed; remove disagreement.

Moment	Data	FIRE benchmark	Disagreement off
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	4.11	1.58
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	4.03	1.54
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.166	0.247
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	−0.169	−0.244	0.546
Fama slope β_F	−0.213	−0.480	1.023
Fama R_F^2	0.0005	0.0135	0.0431
$\rho_1(\Delta\hat{e}_t)$	0.013	−0.050	−0.009

Without disagreement: less relative FX volatility, positive Fama slope and B-S correlation

σ: standard deviation; ρ_1: first-order autocorrelation; $\hat{r}^{(1)}$: one-quarter rate.

▸ all moments

Giving the best chance to the standard model: recalibrating its shocks

Keep FIRE as the benchmark; recalibrate the no-disagreement shock block.

Moment	Data	FIRE benchmark	No disagreement recalibrated shocks
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{c}_t)$	6.43	4.11	2.95
$\sigma(\Delta\hat{e}_t)/\sigma(\Delta\hat{y}_t)$	5.42	4.03	2.94
$\sigma(\hat{r}_t^{(1)} - \hat{r}_t^{(1)*})/\sigma(\Delta\hat{e}_t)$	0.110	0.166	0.309
$\mathrm{Corr}(\Delta\hat{e}_t, \Delta(\hat{c}_t - \hat{c}_t^*))$	−0.169	−0.244	−0.131
Fama slope β_F	−0.213	−0.480	0.906

Recalibration recovers the Backus–Smith sign; the Fama slope remains positive.

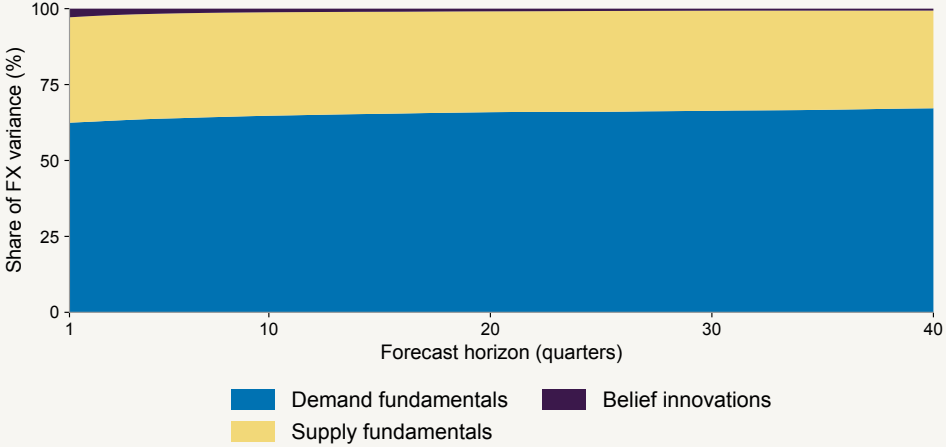
Disagreement matches the data better

σ: standard deviation; $\hat{r}^{(1)}$: one-quarter rate.

▸ all moments

FIRE benchmark: macro shocks drive exchange-rate variation

Demand fundamentals account for 62–67% across horizons. Supply fundamentals second.

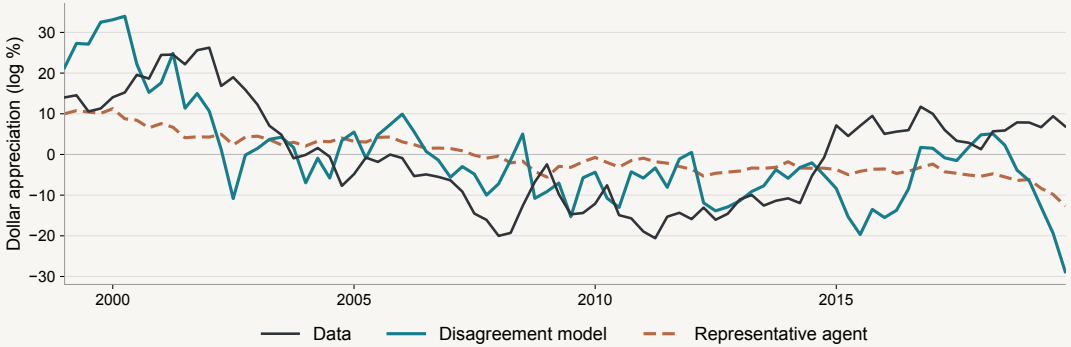


Disagreement matters mainly by amplifying macro shocks.

Reconfirm Kekre-Lenel (2024)'s point: the dominant driver is demand shock

Explaining historical FX volatility: disagreement vs. rep. agent

Infer shocks from yields, output, and disagreement—without using exchange rates.



Dropping disagreement loses most of the historical FX variation.

Conclusion & Outlook

- More in the paper
  - What about a UIP shock? What about average belief bias? Not central
  - Time varying exchange rate volatility caused by disagreement
  - Relevance of survey belief data: CFTC position changes reflect subjective beliefs
- Disagreement among arbitrageurs is **quantitatively central** for exchange-rate dynamics
- Broader implication: FX-macro disconnect partly reflects an **endogenous, difficult-to-observe** UIP wedge generated inside the financial market

Going forward:

- Policy implications
  - Several recent papers explore policy implications given **exogenous UIP wedges**
- Multi-country and NK extensions: special role of the dollar, global financial cycle, etc
- Short and long yield bonds: long bonds tend to be "connected" to FX in the data
- Measurement: hard but several recent papers give some hope
  Giglio et al. (2025); Hacioglu-Hoke et al. (2026)

▸ Time-varying volatility (appendix)

▸ Augmented model (appendix)

Appendix

Literature and contributions

back

Theoretical puzzles and their "solutions"

1. Segmented market risk: Gabaix and Maggiori (2015); **Itskhoki and Mukhin (2021)**; 2. Discount factor shocks: **Kekre and Lenel (2024)**; 3. Convenience yield and liquidity premium: Engel (2016); Valchev (2020)

⇒ Propose a new channel that bridges macro shocks stories and UIP wedges

Expectational errors and exchange rates

Gourinchas and Tornell (2004); Bacchetta and Van Wincoop (2006, 2010); Du, Eusepi and Preston (2021); **Candian and De Leo (2023)**; Molavi, Tahbaz-Salehi, and Vedolin (2024)

⇒ Build a *general equilibrium* international macro model with **heterogeneity** and **subjective beliefs**

Empirical exploration

Lilley, Maggiori, Neiman, and Schreger (2022); Stavrakeva and Tang (2024); Kalemli-Ozcan and Varela (2024); Chahrour, Cormun, De Leo, Guerron-Quintana, and Valchev (2024); Engel and Wu (2024); Miyamoto, Nguyen, and Oh (2025); Fukui, Nakamura, and Steinsson (2025)

⇒ Provide evidence that suggests a connection between exchange rates and disagreement

Heterogeneity and risk premium

Bond market: Xiong and Yan (2010); Barillas and Nimark (2017); Stock market: Caballero and Simsek (2020); Kekre and Lenel (2022); Both: Caramp and Silva (2025)

⇒ Focus on international aspects and exchange rates

Pricing-to-market and incomplete pass-through

back

- Monopolistically competitive final good firms buy the domestic intermediate and sell varieties globally at destination-specific flexible prices
  - They employ no factors and hold no assets: the block exists only to generate incomplete pass-through, as in Kekre and Lenel (2024)
- Elasticity across *imported* varieties moves with the real exchange rate (Alessandria 2009):

$$\theta_{v,t} = \theta_v \left(e_t / \bar{e}\right)^{\xi_v}, \qquad \theta^*_{v,t} = \theta_v \left(e_t / \bar{e}\right)^{-\xi_v}$$

- Markups fall in markets where a firm's relative cost has risen, giving pass-through degree

$$\chi \equiv 1 - \frac{\xi_v}{\theta_v - 1}$$

  - $\xi_v = 0$ restores constant elasticities and the law of one price

Belief formation: idiosyncratic interpretations

back

Opinionated agents, following Caballero and Simsek (2022)

Proposition 1

Same data, unobserved components

$$\hat{x}_t = \tilde{x}_t + \varepsilon_t^x$$

Everyone observes $\hat{x}_t$; the persistent component $\tilde{x}_t$ is hidden.

Each group trusts its own interpretation

$$\hat{x}_t + \mu_t^{xi} =^i \tilde{x}_t + \varepsilon_t^x, \qquad i \in \{A, B\}$$

Group i treats its interpreted signal as unbiased.

$\mu_t^{xi} = \mu^{xi} + \hat{\mu}_t^{xi}$: a constant component plus i.i.d. interpretation shocks.

Why disagreement survives

- Each group regards the other's interpretation as a garbled version of its own.
- Others' beliefs and prices add no information beyond its own interpretation.

Proposition 1: individual beliefs and consensus

Filtered state: $\tilde{x}^i_{t|t} \equiv E^i_t[\tilde{x}_t]$.

G_x: Kalman gain; $\varphi_x = \rho_x(1 - G_x)$: weight on past beliefs.

Individual beliefs: update with the interpreted signal

$$\tilde{x}^i_{t|t} = \varphi_x \tilde{x}^i_{t-1|t-1} + G_x\left(\hat{x}_t + \mu^{xi} + \hat{\mu}^{xi}_t\right)$$

Same data and gain; different interpretations generate different beliefs.

Consensus: average the two groups' beliefs

$$\bar{\tilde{x}}_{t|t} = \varphi_x \bar{\tilde{x}}_{t-1|t-1} + G_x\left(\hat{x}_t + \bar{\mu}^x_t\right)$$

$\bar{\tilde{x}}_{t|t} = \frac{1}{2}(\tilde{x}^A_{t|t} + \tilde{x}^B_{t|t})$, $\quad \bar{\mu}^x_t = \frac{1}{2}(\mu^{xA}_t + \mu^{xB}_t)$.

Zero average bias: consensus follows the rational Kalman filter.

◂ quantitative ◂ analytical ▸ disagreement

Proposition 1: why disagreement is persistent

Disagreement: $d_t^x \equiv \tilde{x}_{t|t}^A - \tilde{x}_{t|t}^B$.

Subtract the two updating equations: the common data cancel

$$d_t^x = (1 - \varphi_x) d_{ss}^x + \varphi_x d_{t-1}^x + G_x \left(\hat{\mu}_t^{xA} - \hat{\mu}_t^{xB} \right)$$

$$d_{ss}^x = \frac{G_x}{1 - \varphi_x} (\mu^{xA} - \mu^{xB}), \qquad \varphi_x = \rho_x (1 - G_x).$$

- Constant differences in interpretation pin down mean disagreement.
- New interpretation differences generate disagreement shocks.
- Past disagreement fades at φ_x: less weight on new data means greater persistence.

Mean-reverting disagreement can coexist with rational consensus beliefs.

◂ quantitative ◂ filtering ▸ higher-order beliefs

Proposition 1: beliefs about others' beliefs

Current beliefs are mutually observed. For $h \geq 1$, group A expects:

Its own forecast to follow the persistent state

$$E_t^A\left[\tilde{x}^A_{t+h|t+h}\right] = \rho_x^h \tilde{x}^A_{t|t}$$

The other group's forecast to move toward its own

$$E_t^A\left[\tilde{x}^B_{t+h|t+h}\right] = \varphi_x^h \tilde{x}^B_{t|t} + (\rho_x^h - \varphi_x^h)\tilde{x}^A_{t|t} - (1-\varphi_x^h) d^x_{ss}$$

Group B reasons symmetrically; the long-run disagreement remains d^x_{ss}.

They disagree about fundamentals, but agree on future disagreement

$$E_t^A[d^x_{t+h}] = E_t^B[d^x_{t+h}] = d^x_{ss} + \varphi_x^h (d^x_t - d^x_{ss})$$

Each group anticipates that the other's excess disagreement will fade.

◂ quantitative ◂ analytical ◂ disagreement

Backus–Smith: the analytical limit

back

As fundamentals and wealth become persistent and feedback approaches its bound,

$$\rho_x, \lambda \to 1, \qquad \Xi \to \Xi_{\max} = 1 + \frac{1}{S}.$$

Negative consumption–FX comovement

$$\mathrm{Corr}(\Delta\hat{c}_t - \Delta\hat{c}_t^*, \Delta\hat{e}_t) \longrightarrow -\sqrt{1 - s_y} < 0$$

$0 \leq s_y < 1$: supply news' share of FX innovation variance.

- Demand shocks lower relative consumption as Home depreciates.
- Amplification strengthens expenditure switching relative to income.

Externally set parameters

back

Parameter	Description	Value	Source
γ	Openness	0.05	Candian & De Leo (2025)
σ	Household risk aversion	5	Candian & De Leo (2025)
θ	Trade elasticity	1.5	Standard
χ	FX pass-through to relative consumer prices	0.600	Kekre & Lenel (2024)
η_A, η_B	Belief group shares	0.5, 0.5	Symmetric groups

- Literature: openness $\in [0.05, 0.15]$, risk aversion $\in [2, 10]$, trade elasticity $\in [0.9, 2]$, FX pass-through $\in [0.5, 0.7]$

FIRE calibration: moment matching

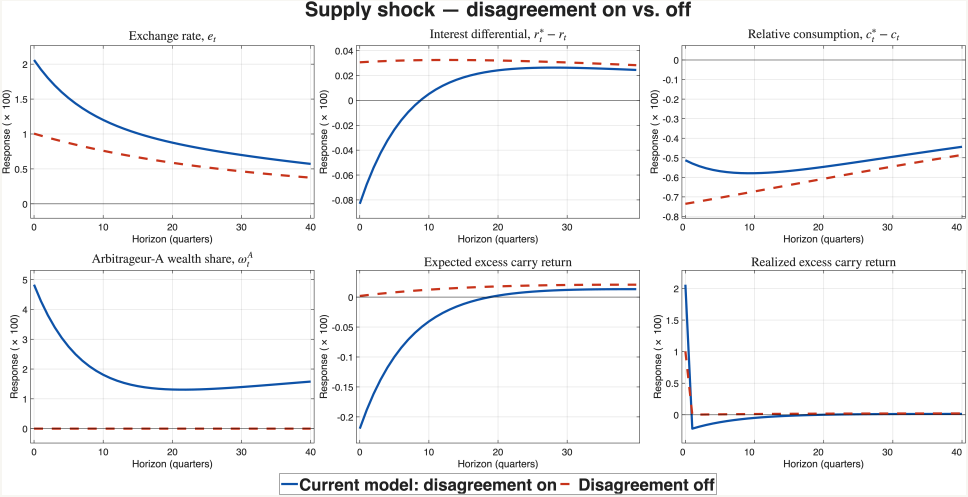
◂ strategy ◂ results

Parameter	FIRE value	Matching moment	Data	FIRE
W	0.07072	Mean intermediary equity/GDP	0.0726	0.0707
ξ	0.009611	AR(1), intermediary equity/GDP	0.922	0.924
Γ	0.003983	SD, quarterly exchange-rate change	3.33%	3.44%
d_{ss}	0.04307	Mean one-quarter disagreement	4.40%	4.31%
G	0.30909	AR(1), one-quarter disagreement	0.648	0.660
$\sigma_{\mu,\beta} = \sigma_{\mu,y}$	0.001782	Innovation SD, disagreement	0.559%	0.584%
ρ_β	0.99010	AR(1), 10-year yield differential	0.940	0.981
σ_β	0.001032	SD, change in annualized 10-year yield differential	0.273%	0.159%
$\mathrm{Corr}(\varepsilon^\beta, \varepsilon^{\beta*})$	0.39290	Correlation, Home and Foreign 10-year yield changes	0.810	0.872
ρ_y	0.98714	AR(1), relative output	0.945	0.944
σ_y	0.008528	SD, relative output growth	0.774%	0.824%
$\mathrm{Corr}(\varepsilon^y, \varepsilon^{y*})$	0.53614	Correlation, Home and Foreign consumption growth	0.629	0.584

Rational average beliefs; heterogeneous beliefs; no UIP shock.

Disagreement amplifies a supply shock

back

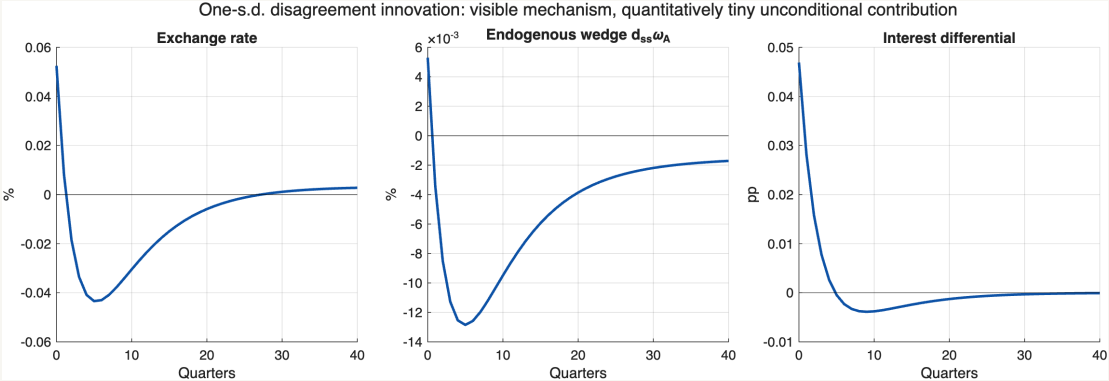


Rate and consumption differentials: Foreign minus Home.

Disagreement-shock IRF (appendix)

Disagreement innovation: dynamics and variance contribution

back



A one-s.d. disagreement innovation moves the wedge and reverses the exchange rate, but disagreement innovations account for only 0.10% of population FX-change variance.

Back to the model IRFs

Key exchange rate moments: full table (A–B)

back

Moment	Data	Augmented-model param.			FIRE	No-disagreement	
		Augmented†	No UIP risk	No avg. bias†	Benchmark	Fixed shocks	Recalib. shocks
A. Exchange Rates and Macro Variables							
$\rho(\Delta e_t)$	0.013	0.064	0.084	-0.043	-0.050	-0.009	-0.003
$\rho(e_t)$	0.950	0.917	0.922	0.900	0.908	0.930	0.926
$\sigma(\Delta e_t)$	0.033	0.040	0.036	0.032	0.034	0.013	0.020
$\sigma(\Delta e_t)/\sigma(\Delta c_t)$	6.43	4.46	4.11	3.58	4.11	1.58	2.95
$\sigma(\Delta e_t)/\sigma(\Delta y_t)$	5.42	4.40	3.98	3.55	4.03	1.54	2.94
$\sigma(\Delta(i_t^{3m,q} - i_t^{3m,q,*}))/\sigma(\Delta e_t)$	0.032	0.426	0.407	0.278	0.117	0.041	0.053
$\sigma(i_t^{3m,q} - i_t^{3m,q,*})/\sigma(\Delta e_t)$	0.110	0.513	0.473	0.374	0.166	0.247	0.309
$\sigma(\Delta(i_t^{10y,q} - i_t^{10y,q,*}))/\sigma(\Delta e_t)$	0.021	0.016	0.013	0.016	0.012	0.055	0.068
$\sigma(i_t^{10y,q} - i_t^{10y,q,*})/\sigma(\Delta e_t)$	0.060	0.061	0.062	0.063	0.074	0.240	0.295
$\mathrm{Corr}(\Delta e_t, \Delta(c_t - c_t^*))$	-0.169	-0.471	-0.320	-0.327	-0.244	0.546	-0.131
B. Excess Returns Predictability							
β^{Fama}	-0.213	-0.449	-0.208	-0.950	-0.480	1.023	0.906
R^2_{Fama}	0.0005	0.0606	0.0184	0.1261	0.0135	0.0431	0.0528
$\beta^{\mathrm{Fama,return}}$	-1.213	-1.449	-1.208	-1.950	-1.480	0.023	-0.094
$R^2_{\mathrm{Fama,return}}$	0.017	0.369	0.252	0.373	0.062	0.024	0.034
$\beta^{\mathrm{Engel,cumul}}_{40q}$	20.65	0.36	1.19	-2.10	-6.52	-3.77	-5.72
90% interval	[6.24, 35.06]	[-3.08, 3.99]	[-2.43, 4.65]	[-6.51, 2.63]	[-18.54, 3.27]	[-36.67, 26.99]	[-35.43, 21.28]

†With UIP shock. FIRE and augmented models are separately calibrated.

No-disagreement: FIRE structure, fixed or recalibrated macro shocks.

panels C–D

Key exchange rate moments: full table (C–D)

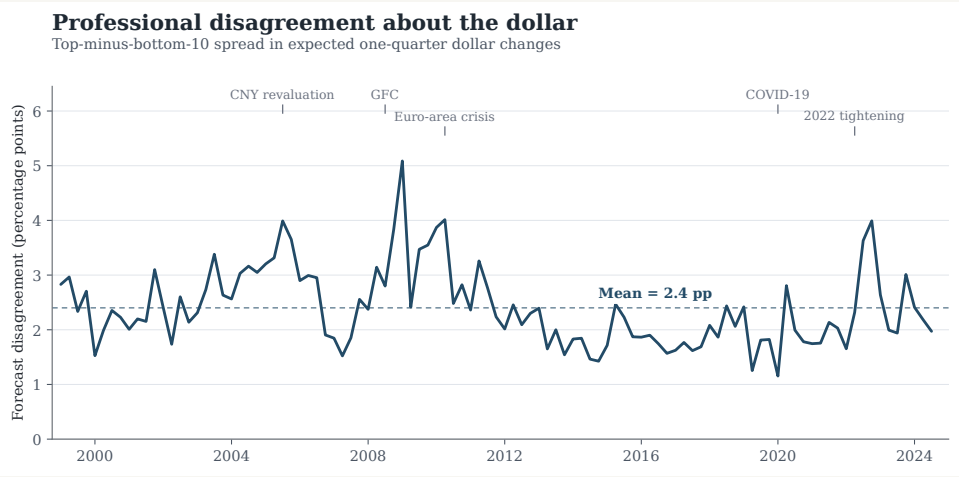
back

Moment	Data	Augmented-model param.			FIRE	No-disagreement	
		Augmented†	No UIP risk	No avg. bias†	Benchmark	Fixed shocks	Recalib. shocks
C. Changes-on-Changes Relationships							
Δe_t on $\Delta(r_t^{(1q)} - r_t^{(1q),*})$	-0.50	-0.13	-0.33	0.56	1.62	-4.10	-2.90
90% interval	[-2.03, 1.03]	[-0.21, -0.04]	[-0.41, -0.26]	[0.43, 0.68]	[0.35, 3.38]	[-4.78, -3.44]	[-3.43, -2.36]
Adjusted R^2	-0.005	0.046	0.283	0.381	0.358	0.444	0.364
Δe_t on $\Delta(r_t^{(10y)} - r_t^{(10y),*})$	-1.42	1.24	-6.82	2.57	-15.03	-4.16	-3.57
90% interval	[-3.37, 0.52]	[-1.75, 4.08]	[-10.70, -2.84]	[-0.28, 5.28]	[-18.02, -11.13]	[-4.45, -3.85]	[-3.72, -3.41]
Adjusted R^2	0.005	0.010	0.124	0.030	0.487	0.825	0.928
D. K–L-Style Level Relationships							
q_t on $r_t^{(1q),*} - r_t^{(1q)}$	-3.47	-0.22	-0.45	0.34	0.55	-1.79	-1.23
90% interval	[-6.81, -0.12]	[-0.55, 0.07]	[-0.80, -0.13]	[-0.22, 0.83]	[-1.48, 2.42]	[-4.06, 0.24]	[-2.60, 0.19]
Adjusted R^2	0.222	0.038	0.104	0.084	0.084	0.344	0.276
q_t on $r_t^{(10y),*} - r_t^{(10y)}$	-7.79	1.98	0.42	-2.75	-6.65	-2.23	-1.70
90% interval	[-12.81, -2.77]	[-4.70, 8.96]	[-6.72, 7.90]	[-7.94, 3.44]	[-10.66, -2.80]	[-4.27, -0.24]	[-3.09, -0.21]
Adjusted R^2	0.336	0.126	0.111	0.169	0.521	0.471	0.448

†With UIP shock. Data intervals: 90% confidence intervals.

Model intervals: 5th–95th percentiles across simulations.

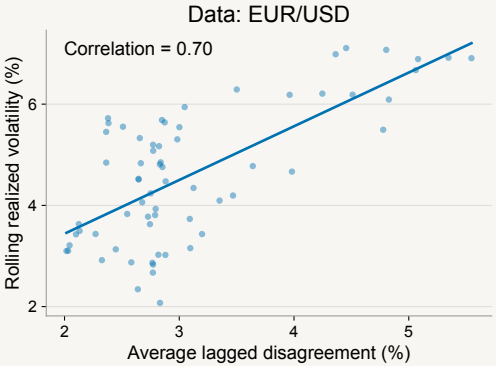
Disagreement over time

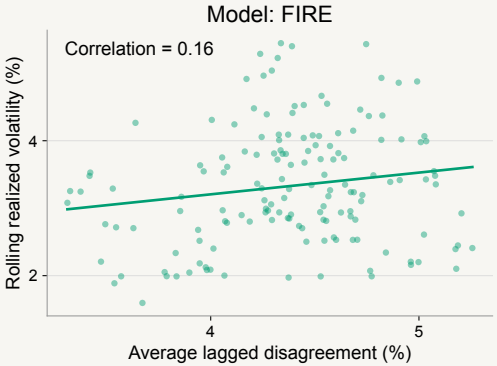


Source: Blue Chip Financial Forecasts; one-quarter dollar forecasts.

Time-varying and persistent; the gap never closes in the sample.

Disagreement and time-varying exchange-rate volatility



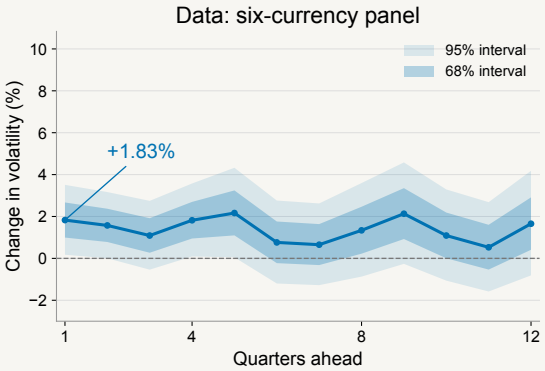


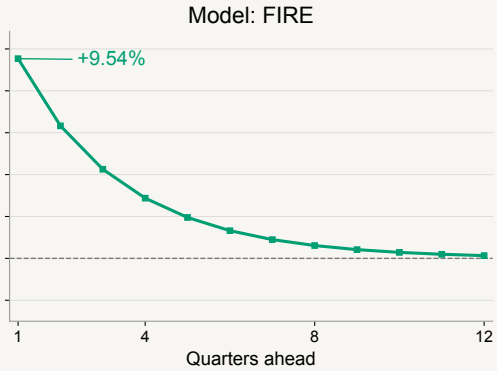
More disagreement goes with greater exchange-rate volatility.

Both panels: 8-quarter SD of quarterly FX changes vs. average lagged disagreement.

▸ monthly by currency ▸ measurement ▸ responses ◂ conclusion

Disagreement predicts persistent exchange-rate risk





Greater disagreement raises exposure to subsequent exchange-rate movements.

Data: one residual SD of disagreement; 68% and 95% pointwise intervals. Local projection with controls. Model: one-SD disagreement shock.

◂ scatter ▸ specification ◂ conclusion

Disagreement and volatility: measurement and specification

Data

- Disagreement: SD of 90-day FX forecasts / current spot; at least eight forecasters.
- Scatter: 8-quarter SD of quarterly FX changes. Response: daily-based realized volatility.
- Quarterly projections: six currencies, currency and quarter effects; controls for risk, disagreement, FX changes, and forecaster counts.

Model

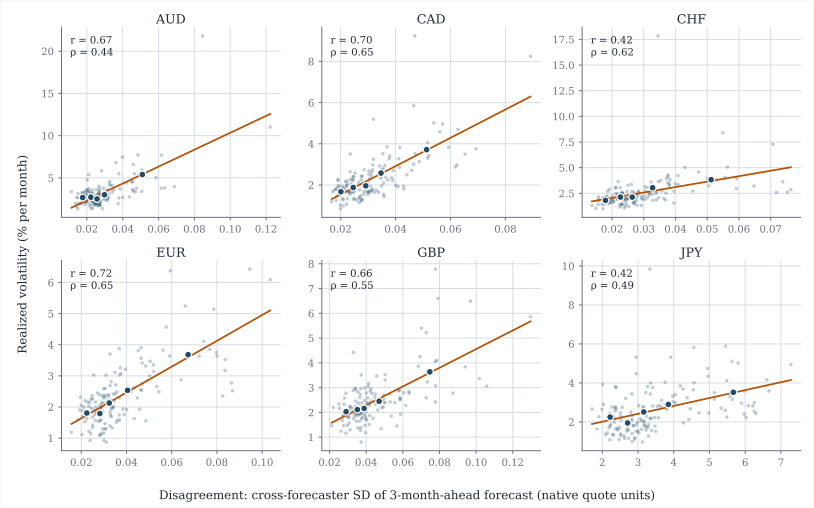
- FIRE benchmark with both cross-terms; no UIP shock.
- Scatter: the same rolling-window construction as the data.
- Response: conditional FX volatility after a one-SD disagreement innovation.

Data provide predictive evidence, not an externally identified causal effect. Model and data shock sizes are normalized separately.

◂ scatter ◂ responses

Disagreement vs. realized FX volatility: scatter

back



Quintile means and OLS fit; native quote units per panel.

Augmented model: average belief bias and UIP shock

Add average belief bias and a UIP shock to the FIRE benchmark.

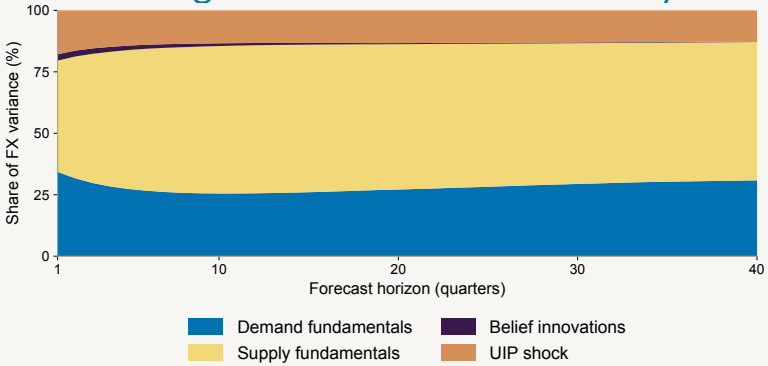
- Introduce shock misperception in the same spirit as Candian-De Leo (2025)
- Reintroduce a reduced-form UIP shock
  - Its dynamics are disciplined by a risk measure (excess bond premium or VIX) as proposed by Kekre-Lenel (2024)
- Average bias improves the "predictability reversal" moments: Engel (2016)
- When externally disciplined, the UIP shock remains a minor driver of exchange-rate variation
  - Up to 18%

These ingredients help match certain moments,
but the FIRE benchmark already captures the main disconnect.

shock decomposition

conclusion

Reintroducing the UIP shock: a secondary source of FX variation



80–87%
Macro fundamentals

13–18%
UIP shock

Shares across 1–40 quarters.

At fixed augmented parameters, adding the UIP shock raises short-rate/FX volatility:

$$0.473 \longrightarrow 0.513 \qquad \text{(data: } 0.110\text{)}$$

The FIRE benchmark's central results do not require this shock.

overview